

POWER BI – DATA MODELING & VISUALIZATION DOCUMENTATION

Global Female Primary Education Completion Analysis

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Course: Program In AI Driven Data Analytics

Batch: DA62

Project Type: Mini Project

Data Source: World Bank – Primary completion rate, female (% of relevant age group)

1. Project Overview

The **Global Female Primary Education Completion Analysis** project analyzes female primary education completion rates across countries and years using World Bank data.

Power BI was used for data modeling, relationship creation, DAX calculations, and interactive data visualization.

The main focus of the project is to understand completion-rate trends over time and compare completion rates across countries and available geographical and income-group categories.

2. Importing Data

The cleaned datasets were imported into **Power BI Desktop** for data modeling, analysis, and visualization.

The following tables were imported:

- **Primary_Completion_Rate**
 - **Country_Master**
 - **Country_Metadata**
 - **Completion_Rate_Metadata**
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3. Creating Relationships

Relationships were created between the tables using common key fields.

3.1 Country Relationship

Country_Master[Country_Code] → Primary_Completion_Rate[Country_Code]

- Cardinality: One-to-Many (1:*)

- Active relationship
- Connects country information with completion-rate records.

3.2 Indicator Relationship

Completion_Rate_Metadata[Indicator Code] → Primary_Completion_Rate[Indicator_Code]

- Cardinality: One-to-Many (1:*)
- Active relationship
- Connects indicator and source information with completion-rate data.

3.3 Country Metadata Relationship

Country_Master[Country_Code] → Country_Metadata[Country_Code]

- Cardinality: One-to-Many (1:*)
- Active relationship
- Connects country information with additional metadata.

4. DAX Operations

After completing data cleaning and data modeling, **DAX (Data Analysis Expressions)** was used to create calculations and measures for analysis and dashboard visualizations.

4.1 Average Completion Rate

Average Completion Rate =

AVERAGE(Primary_Completion_Rate[Completion_Rate])

Purpose: Calculates the average female primary education completion rate across the available records.

4.2 Number of Countries

Number of Countries =

DISTINCTCOUNT(Country_Master[Country_Code])

Purpose: Counts the number of unique countries in the dataset.

4.3 Maximum Completion Rate

Maximum Completion Rate =

MAX(Primary_Completion_Rate[Completion_Rate])

Purpose: Identifies the highest completion-rate value recorded in the dataset.

4.4 Minimum Completion Rate

Minimum Completion Rate =

MIN(Primary_Completion_Rate[Completion_Rate])

Purpose: Identifies the lowest completion-rate value recorded in the dataset.

4.5 Total Records

Total Records =

COUNTROWS(Primary_Completion_Rate)

Purpose: Counts the total number of records available in the Primary_Completion_Rate table.

4.6 Number of Regions

Number of Regions =

DISTINCTCOUNT(Country_Metadata[Region])

Purpose: Counts the unique geographical regions available in the metadata.

4.7 Number of Income Groups

Number of Income Groups =

DISTINCTCOUNT(Country_Metadata[IncomeGroup])

Purpose: Counts the distinct income-group categories available in the metadata.

4.8 Completion Rate Range

Completion Rate Range =

IF (

[Completion Rate] < 20,

"0–20%",

IF(

[Completion Rate] < 40,

"20–40%",

IF(

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[Completion Rate] < 60,
"40–60%",
IF(
    [Completion Rate] < 80,
    "60–80%",
    IF(
        [Completion Rate] < 100,
        "80–100%",
        "100%+"
    )
)
)
)
)
)
)

```

Purpose: Categorizes completion-rate values into different percentage ranges. This calculation was used for completion-rate range analysis and the corresponding **Clustered Column Chart**.

These DAX calculations were created to support the analysis of **female primary education completion rates** across countries and years. The calculations were used in **Cards and other Power BI visuals**.

5. Visualizations

The following Power BI visuals were created:

- **Cards** – Display key summary values such as average, maximum, and minimum completion rates, number of countries, and total records.
- **Gauge Chart** – Represents completion-rate performance against a reference or target value.
- **Line Chart** – Shows completion-rate trends across the available years.
- **Clustered Column Charts** – Used for region-wise comparison and completion-rate range analysis.

- **Map** – Displays country-level completion-rate information geographically.
 - **Matrix** – Presents completion-rate information in a structured format.
 - **Donut Chart** – Shows year-wise completion-rate information on the detailed analysis page.
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6. Slicers and Interactivity

Slicers were added to make the dashboard interactive.

The main slicers used were:

- **Country**
- **Year**

These slicers allow users to filter the available data and dynamically update the connected visuals.

A **Year selection list** was also added to the detailed analysis page.

7. Drill-through Analysis

A separate detailed analysis page was created using the **Drill-through** feature.

- **Country** was added as the drill-through field.
- Users can right-click a country or relevant data point and select **Drill through**.
- The detailed page displays information based on the selected country.

This allows users to move from the overall dashboard to detailed country-level analysis.

8. Dashboard Formatting

The dashboard was formatted to improve readability and presentation.

The following formatting was applied:

- Appropriate titles were added to the visuals.
- X-axis and Y-axis titles were added where applicable.
- Visual sizes and positions were adjusted.
- Text and value sizes were adjusted for readability.
- Visuals were arranged into a clear dashboard layout.

- Interactive elements were organized for easier navigation.
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9. Data Limitation

The dataset provided detailed country-level completion-rate information. However, the additional **region and income-group metadata was limited** and did not provide complete information for all countries and categories.

Therefore, regional and income-group analysis was performed only using the available metadata. These dimensions were not treated as complete global comparisons.

The main focus of the dashboard is **country-level and year-wise female primary education completion-rate analysis**.

10. Key Analysis

The dashboard allows users to:

- View overall completion-rate statistics.
 - Compare completion rates across countries.
 - Analyze completion-rate trends over the available years.
 - Understand the distribution of completion-rate values.
 - Analyze countries based on completion-rate ranges.
 - View country-level completion-rate information on a map.
 - Compare available regional and income-group information.
 - Filter the dashboard using country and year slicers.
 - Drill through to detailed country-level analysis.
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11. Project Summary

The **Global Female Primary Education Completion Analysis** project uses World Bank data to analyze female primary education completion rates across countries and years.

Power BI was used for **data modeling, relationship creation, DAX calculations, and interactive visualization**.

The dashboard provides country-level and time-based analysis, with additional regional and income-group analysis where sufficient metadata is available. A drill-through feature was also implemented to provide detailed country-level analysis.

12. Tools Used

Power BI Desktop | Power Query | DAX | Microsoft Excel

13. Conclusion

The analysis provides an overview of female primary education completion rates across countries and regions over the available years.

The dashboard shows that **female primary education completion has generally improved over time**, indicating positive long-term progress in primary education completion for girls.

The analysis also highlights differences in completion rates across **countries, regions, and income groups**. While several countries show relatively high completion rates, some countries have comparatively lower rates, indicating areas where further improvement may be needed.

Overall, the findings suggest a **positive long-term trend in female primary education completion**, while also showing that progress is not uniform across all countries and regions.